GamersDen - A Smart Platform for Gamers and Gaming Centres

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Abstract—

GamersDen is a web-based platform designed to connect customers with nearby PlayStation gaming centres and simplify gaming-session discovery and booking. Customers may need to contact several venues manually to check location, facilities, prices and available time slots. GamersDen proposes a centralized system where gaming centres can publish venue information and manage availability, while customers can search for suitable nearby centres and reserve a preferred slot. The frontend is built using HTML, CSS, Java and React, the backend uses Django for application logic and request handling, and MySQL provides persistent relational storage. The database is designed to maintain user, gaming-centre, facility, slot and booking records. The booking workflow validates availability before saving a reservation, helping reduce conflicting bookings and manual communication. GamersDen aims to make gaming-centre discovery more convenient for customers and provide venue operators with a structured way to manage reservations. This paper describes the system motivation, architecture, database design, workflow, evaluation plan, challenges and future enhancements.

Index Terms— PlayStation booking, gaming centres, location-based services, React, Django, MySQL, online reservation.

I. INTRODUCTION

Gaming and esports centres provide social entertainment spaces where customers can reserve consoles and play with friends. However, information about venues, prices, facilities and availability may be scattered across social-media pages, messaging applications and direct phone calls. This can make discovery time-consuming and leave customers uncertain about whether a preferred slot is available. GamersDen is proposed as a centralized platform connecting customers with nearby PlayStation gaming centres. Customers can explore venue details and available sessions, while operators can maintain venue information and manage bookings. The system combines a responsive web interface with a Django backend and MySQL database.

A. Problem Statement

Manual booking methods do not always provide a consistent view of venue availability and can require customers to contact multiple centres. Operators may also face difficulties maintaining reservations and preventing duplicate bookings. A unified web-based system is needed to organize venue discovery, slot availability and reservations.

B. Objectives

- Provide a centralized platform to discover nearby PlayStation gaming centres.
- Display venue details, facilities, pricing and available booking slots.
- Allow customers to create and review reservations online.
- Help venue operators manage venue information, availability and bookings.
- Reduce manual communication and improve booking transparency.
- Implement the application using React, Django and MySQL.

C. Research Gap

Venue information is often distributed across individual websites and social platforms, while general reservation systems may not represent gaming-specific facilities and session slots. GamersDen focuses on combining proximity-oriented venue discovery with gaming-session booking and operator-side management in one platform.

II. LITERATURE SURVEY

Location-based services provide a foundation for finding services near a user's chosen location. Online reservation systems demonstrate the value of centralized availability and booking records. Gaming and esports platforms support digital communities and gaming-related services, while GamersDen focuses specifically on physical PlayStation-centre discovery and time-slot reservation.

React supports component-based interfaces and interactive user experiences. Django provides server-side request handling, application logic and authentication facilities. MySQL is a relational database suitable for organizing users, venues, slots and booking records. These technologies form the technical foundation of the proposed system.

The design draws on established concepts in location-based services, web application architecture, database management and software engineering. The contribution of GamersDen is the application of these concepts to a gaming-centre booking workflow.

III. SYSTEM ARCHITECTURE

A. Architecture Diagram

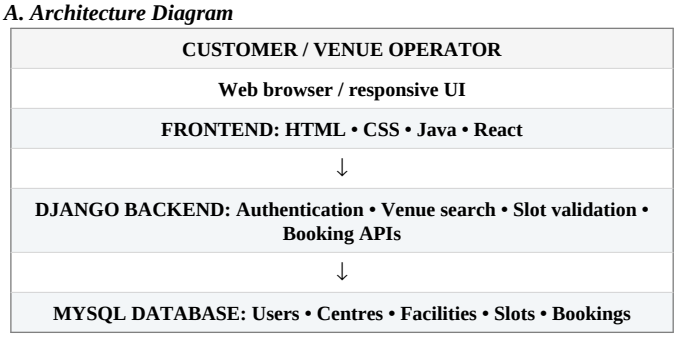

Fig. 1. High-level architecture of GamersDen.

B. Architecture Description

GamersDen follows a three-layer architecture. The presentation layer uses HTML, CSS, Java and React to provide customer and operator interfaces. The application layer uses Django to handle authentication, venue search, availability checks, booking validation and API requests. The data layer uses MySQL to store user accounts, gaming-centre details, facilities, time slots and bookings. Separating these layers supports maintainability and future expansion.

C. Booking Workflow

- **1) Registration and login:** Customers and authorized venue operators access the platform through their accounts.
- **2) Venue discovery:** The customer searches by location or selects a location to find nearby gaming centres.
- **3) Venue details:** The customer reviews facilities, pricing, contact information and available sessions.
- **4) Slot selection:** The customer selects a suitable date and time slot.
- **5) Availability validation:** Django checks the slot status before accepting the reservation.
- **6) Booking confirmation:** A valid booking is stored in MySQL and linked to the customer, venue and slot.
- **7) Operator management:** Venue operators review reservations and update relevant venue or availability information.

D. Database Design

The database may contain a User table for customer and operator accounts; a GamingCentre table for venue name, address, location and contact details; a Facility or Console table for equipment information; a Slot table for date, time, duration and availability; and a Booking table linking the customer to the selected venue and slot. Primary keys and foreign-key relationships support referential integrity. Booking validation should ensure that a slot cannot be reserved more than once for overlapping sessions.

E. Technology Stack

Frontend: HTML, CSS, Java and React. Backend: Django for server-side logic and API handling. Database: MySQL for persistent relational storage. The frontend communicates with Django through application endpoints, and Django reads or updates records in MySQL.

F. Security and Validation

The application should validate input on both client and server sides. Authentication and authorization should restrict venue-management functions to approved operator accounts. Booking operations should re-check availability before saving records. Passwords must be securely hashed, and database operations should use safe ORM or parameterized mechanisms.

IV. EVALUATION PLAN AND DISCUSSION

A. Evaluation Metrics

Criterion	Measure	Purpose
Venue discovery	Search response and relevant nearby results	Discovery convenience
Booking reliability	Successful bookings without conflicts	Reservation accuracy
Availability consistency	Correct slot state after booking	Data consistency
Response time	Response time for key operations	Performance
Usability	Feedback on navigation and booking	User experience

TABLE I. PLANNED EVALUATION OF THE GAMERSDEN PLATFORM.

B. Discussion

Testing should verify registration, login, venue search, venue-detail display, slot availability, booking creation and operator-side management. Repeated or concurrent booking attempts can be used to test slot consistency. Response time can be measured for venue search and booking operations, while user feedback can assess navigation and ease of use. No numerical results are claimed in this paper; these are proposed evaluation criteria.

C. Comparison with Existing Approaches

Manual booking commonly relies on calls or messages and may not provide a unified availability view. General listings can help customers find venues but may not support gaming-specific slot reservation. GamersDen is designed to combine nearby venue discovery, gaming-centre information, slot selection and structured booking records.

V. CHALLENGES FACED

- Maintaining accurate and up-to-date slot availability for each venue.
- Preventing duplicate reservations when customers select the same slot.
- Representing venue location and proximity clearly.
- Designing consistent relationships among users, venues, slots and bookings.
- Supporting responsive layouts on mobile and desktop devices.
- Protecting user accounts and reservation information.

VI. CONCLUSION AND FUTURE WORK

GamersDen proposes a web-based approach for connecting customers with nearby PlayStation gaming centres and enabling structured session booking. The React-based frontend, Django backend and MySQL database support venue discovery, availability management and reservation records. By bringing these functions into one platform, the system aims to reduce dependence on manual communication and improve convenience for customers and operators.

Future enhancements may include map integration and distance-based ranking, online payments, automated notifications, ratings and reviews, promotions, owner analytics, cancellation workflows and mobile applications. The platform could also be extended to other gaming and recreational venues.

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